Do Now · retrieval

Lesson 101 · Multicore CPUs and Parallel Systems

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a core?

Q2

Define concurrency.

Q3

What does Amdahl's law illustrate?

Q4

A 3×4 array of independent cells: how many cells?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — A multicore CPU integrates two or more processing cores, which can execute independent threads concurrently.

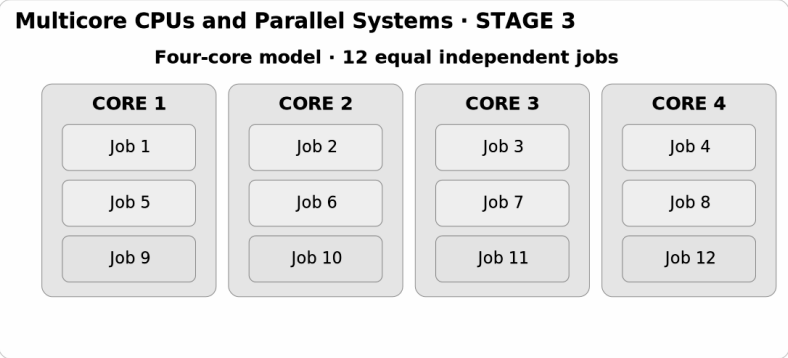
Core idea 2

Apply — Parallel processing can reduce completion time when tasks can be divided; a serial dependency is a critical-path constraint.

Core idea 3

Evaluate — Shared memory bandwidth, locking, thread scheduling and the serial proportion of a program limit speedup. More cores do not guarantee proportional gains.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

multicore — Processor with multiple CPU cores.

parallelism — Simultaneous processing of independent tasks.

critical path — Longest dependent chain.

Amdahl's law — Speedup bounded by serial portion.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

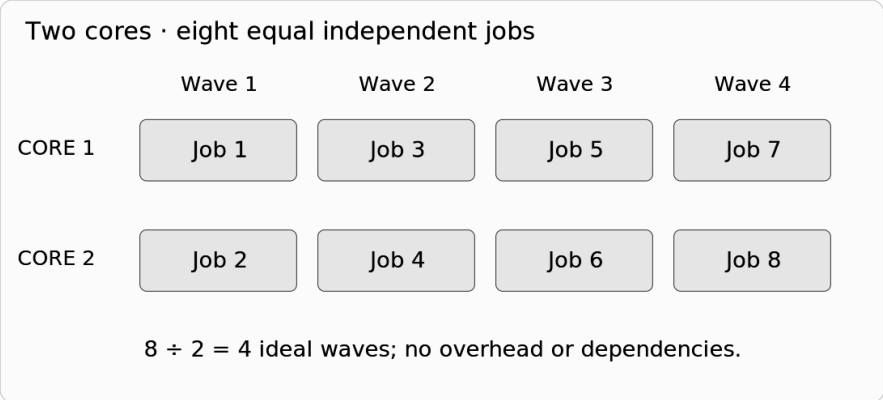
Problem

Eight equal independent jobs are shared across two identical cores. How many ideal waves?

Follow the worked method

1. Explain the choice — 8 ÷ 2 = 4 equal-duration waves, versus 8 on one core.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the choice
12 ÷ 4 = 3 waves; shared-memory contention or scheduling overhead can extend time.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Twelve equal independent jobs run on three identical cores. Compare ideal waves with one core. [6]

Question 2

Explain two benefits and two limitations of multicore processing for video rendering. [8]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Half a program must run serially; other half parallelised perfectly across four cores. Compute ideal speedup and interpret. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Four CPU cores always give exactly four times the performance of one core.

Use key vocabulary from memory

1. Explain multicore in your own words.

2. Explain parallelism in your own words.

3. Explain critical path in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.